



**CALICUT UNIVERSITY – FOUR-YEAR
UNDERGRADUATE PROGRAMME (CU-FYUGP)
BSc PHYSICS HONOURS**

Programme	B. Sc. Physics Honours				
Course Title	FOUNDATIONS OF DATA SCIENCE				
Type of Course	Major Elective (SPECIALIZATION IV: DATA SCIENCE AND ARTIFICIAL INTELLIGENCE)				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Fundamental Programming Concepts in Python 2. Basic idea of linear algebra				
Course Summary	The course will introduce the fundamental concepts of linear algebra, probability and statistics required for a program in data science				

Course Outcomes (CO):

CO	CO Statement	Cognitive level*	Knowledge Category#	Evaluation Tools used
CO1	Students will evaluate eigenvalues and eigenvectors to decompose matrices, enabling them to analyze and interpret data transformations effectively	An	P	Instructor-created exams / assignment
CO2	Proficiency in solving linear equations using linear algebra and understanding the geometric interpretation of solutions.	Ap	P	Instructor-created exams / assignment
CO3	Students will apply fundamental probability concepts to solve real-world problems	Ap	P	Assignment / Quiz

CO4	tudents will utilize statistical techniques for data interpretation and decision-making.	Ap	P	structor-created exams / assignment
CO5	tudents will apply sampling techniques and hypothesis tests to make inferences about populations from sample data, using one-tailed, two-tailed tests, and ANOVA for analysis.	Ap	C	assignment / Case Studies
CO6	develop critical thinking and problem-solving skills	E	M	Assignment / Case Studies
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create(C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Mark
I	Linear Algebra		10	12
	1	Matrices: Properties of matrix, Various kind of Matrices	1	
	2	Elementary Transformations of Matrices and Rank of Matrices	2	
	3	Determinants, Minors, Cofactors, Inverse of a matrix	2	
	4	Linear Independence: Characteristic equations, Eigen values and Eigen Vector	2	
	5	Solving system of linear equations: Gauss Elimination Method, Gauss Jordan method	3	
	Relevant sections of Book 1			
I I	Basic Statistics and Descriptive Measures		10	15
	6	Measures of Central Tendency	2	
	7	Measures of Dispersion	2	
	8	Measures of Skewness	2	
	9	Measures of Kurtosis	2	
	10	Correlation and Regression	2	
	Relevant sections of Book 2			
III	Theory of Probability		13	21
	11	Classical and Empirical Probability	2	

	12	Events, Algebra of events	2	
	13	Classical approach to probability, Axiomatic definitions of probability, Simple problems	2	
	14	Theorems of probability - Addition Theorem, Multiplication Theorem	2	
	15	Conditional probability	2	
	16	Bayes' Theorem and Geometrical Probability – Examples and Problems	3	
	Relevant sections of Book 2			
I V	Advanced Probability Distributions		15	22
	17	Discrete and continuous random variables and probability distribution	2	
	18	Binomial distribution: Definition, Expectation, Variance, Moment Generating Function and Problems	3	
	19	Poisson distribution: Definition, Expectation, Variance, Moment Generating Function and Problems	3	
	20	Normal distribution: Definition, Expectation, Variance, Moment Generating Function, Standard normal curve and Problems	3	
	21	Testing of Hypothesis: General principles of testing, Two types of errors	2	
	22	Type of Testing: T-Test, ANOVA-Test, Chi-square test (Basics only)	2	
	Relevant sections of Book 2			
V	Open Ended Module		12	
	Books and References: 1. Gilbert Strang, "Introduction to Linear Algebra", Wellesley-Cambridge Press (Book 1) 2. Fundamentals of Mathematical Statistics. S.C. Gupta , V.K. Kapoor, Sultan Chand & Sons, 2020 (Book 2) 3. Introduction to Mathematical Statistics, Hogg R V Craig A T, Macmillan 4. Probability and Statistics for Engineers, Miller I Freund J E, Prentice Hall of India 5. Advanced Engineering Mathematics, Erwin Kreyszig, Wiley 6. Higher Engineering Mathematics, B S Grewal, Khanna Publishers			

Mapping of COs with PSOs and POs :

	PSO 1	PSO-2	PSO 3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	1	2	2	3	1	3	1	2	2	3	2	1
CO 2	3	1	2	2	3	1	3	1	2	2	3	2	1
CO 3	2	2	3	2	2	2	3	1	3	2	3	2	2
CO 4	2	2	3	2	2	2	3	1	3	2	3	2	2
CO 5	2	2	3	2	2	2	3	1	3	2	3	2	2
CO 6	1	1	2	2	1	2	3	3	2	2	3	3	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	